

```

In[*]:= If[! TrueQ[Private`$ev], Block[{Private`$ev = True},
  Print["Total time: ", First@AbsoluteTiming[ClearAll["Global`*"];
    NotebookEvaluate[EvaluationNotebook[], InsertResults -> True]], " s"]]]

(* =====*)
(*Verification of low Mach number expansion for u and φ up to 8th order*)
(*All equations should yield {0,0} or 0. If all pass,prints True.*)
(* =====*)

(*_ 0th order _*)
u0 = {(3 Tf) / (4 (3 μ + α)) (2 μ + α - α x2^2), 0};

(*_ 2nd order _*)
φ2 = (3 Tf (2 - x1) α μ) / (2 (α + 3 μ));

u2 = {(9 Tf^2 α (Tf x2^4 α ((-5 + x2^2) α - 10 μ) + 40 (-2 + x1) μ (α - x2^2 α + 2 μ) (α + 3 μ))) /
  (320 (α + 3 μ)^3), 0};

(*_ 4th order _*)
φ4 =
  - ((9 Tf^2 α μ (3 Tf x1 (α + 2 μ)^2 + 4 (α + 3 μ) ((-1 + (-4 + x1) x1 + x2^2) α μ + (-1 + x2^2)
    α ν - 2 μ (μ + ν)))) / (32 (α + 3 μ)^3));

u4 = {- (1 / (179 200 (α + 3 μ)^5)) 27
  Tf^3 α (Tf^2 x2^4 α ((495 + 35 x2^2 - 75 x2^4 + 8 x2^6) α^3 +
    10 (308 + 14 x2^2 - 15 x2^4) α^2 μ + 140 (45 + x2^2) α μ^2 + 4200 μ^3) -
    40 Tf μ (α + 3 μ) (3 (39 x1 - 35 x1 x2^2 - 35 (-2 + x1) x2^4 + 7 (-2 + x1) x2^6) α^3 +
    14 (30 x2^4 + x1 (47 - 15 x2^2 (2 + x2^2))) α^2 μ -
    420 x1 (-3 + x2^2) α μ^2 + 840 x1 μ^3) +
    2800 μ (α + 3 μ)^2 (μ ((-14 - 6 (-4 + x1) x1 + 6 (2 + (-4 + x1) x1) x2^2 - x2^4) α^2 -
    4 (3 (-4 + x1) x1 + 2 (3 + x2^2)) α μ + 8 μ^2) + 2 (α + 2 μ) (α - 2 x2^2 α + 2 μ) ν)),
  - ((27 Tf^4 x2 α^3 μ ((6 - 7 x2^4 + x2^6) α - 14 (-1 + x2^4) μ)) / (2240 (α + 3 μ)^4))};

(*_ 6th order _*)
φ6 = 1 / (17 920 (α + 3 μ)^5) × 27 Tf^3 α μ
  (-56 Tf x2^4 α^2 ((-5 + x2^2) α - 10 μ) μ (α + 3 μ) - 4 (α + 3 μ) (840 (-2 + x1) α
    ((-1 + x2^2) α - 2 μ) μ (α + 3 μ) + 7 Tf (- (15 - 15 x2^2 - 5 x2^4 + x2^6) α^3) +
    10 (-9 + 6 x2^2 + x2^4) α^2 μ + 60 (-3 + x2^2) α μ^2 - 120 μ^3)) (μ + ν) +
  x1 (-1680 Tf (-3 + x1) α μ (α + 2 μ)^2 (α + 3 μ) - 3 Tf^2 (α + 2 μ)
    (117 α^3 + 658 α^2 μ + 1260 α μ^2 + 840 μ^3) +
    1120 α μ (α + 3 μ)^2 (- (3 + (-6 + x1) x1) α μ + 2 α ν + 2 μ (5 μ + 2 ν))));

u6 = {1 / (2 870 067 200 (α + 3 μ)^7) × 27 Tf^4 α
  (3 843 840 Tf x1^2 α^3 μ^2 (α + 3 μ)^2 ((6 - 35 x2^4 + 7 x2^6) α + 14 (1 - 5 x2^4) μ) - 1456
    x1 α^2 μ (α + 3 μ) (10 560 Tf α μ (α + 3 μ) ((6 - 35 x2^4 + 7 x2^6) α + 14 (1 - 5 x2^4) μ) +
    Tf^2 ((-9530 + 50 985 x2^4 - 1155 x2^6 - 4950 x2^8 + 528 x2^10) α^3 -
    44 (1367 - 7035 x2^4 + 105 x2^6 + 225 x2^8) α^2 μ -
    660 (188 - 945 x2^4 + 7 x2^6) α μ^2 + 83 160 (-1 + 5 x2^4) μ^3) -
    36 960 (-1 + 5 x2^4) α μ (α + 3 μ)^2 (3 μ + 2 ν)) + 24 024 (2 - x1) α
    μ (α + 3 μ) (Tf^2 x2^4 α ((495 + 35 x2^2 - 75 x2^4 + 8 x2^6) α^3 +

```

$$\begin{aligned}
& 10 (308 + 14 x^2 - 15 x^4) \alpha^2 \mu + 140 (45 + x^2) \alpha \mu^2 + 4200 \mu^3 - \\
& 40 T_f \mu (\alpha + 3 \mu) (3 (39 x - 35 x^2 - 35 (-2 + x) x^2 + 7 (-2 + x) x^6) \alpha^3 + \\
& 14 (30 x^4 + x (47 - 15 x^2 (2 + x^2))) \alpha^2 \mu - \\
& 420 x (-3 + x^2) \alpha \mu^2 + 840 x \mu^3) + 2800 \mu (\alpha + 3 \mu)^2 \\
& (\mu ((-14 - 6 (-4 + x) x + 6 (2 + (-4 + x) x) x^2 - x^4) \alpha^2 - 4 (3 (-4 + x) x + \\
& 2 (3 + x^2)) \alpha \mu + 8 \mu^2) + 2 (\alpha + 2 \mu) (\alpha - 2 x^2 \alpha + 2 \mu \nu)) + \\
& 840 840 \alpha \mu (\alpha + 3 \mu) (T_f x^4 \alpha ((-5 + x^2) \alpha - 10 \mu) + 40 (-2 + x) \mu \\
& (\alpha - x^2 \alpha + 2 \mu) (\alpha + 3 \mu)) (3 T_f x (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) ((-1 + (-4 + x) x + x^2) \alpha \mu + (-1 + x^2) \alpha \nu - 2 \mu (\mu + \nu))) + \\
& x^2 \alpha (192 192 T_f^2 \alpha \mu (\alpha + 3 \mu) ((495 + 35 x^2 - 75 x^4 + 8 x^6) \alpha^3 + \\
& 10 (308 + 14 x^2 - 15 x^4) \alpha^2 \mu + 140 (45 + x^2) \alpha \mu^2 + 4200 \mu^3) + \\
& T_f^3 ((-4 403 035 - 1918 917 x^2 + 1301 157 x^4 + 101 101 x^6 - 70 616 x^8 + \\
& 5067 x^{10}) \alpha^5 - 91 (497 222 + 180 312 x^2 - 87 813 x^4 - 4444 x^6 + 1552 \\
& x^8) \alpha^4 \mu + 4004 (-46 230 - 12 894 x^2 + 4050 x^4 + 101 x^6) \alpha^3 \mu^2 + \\
& 360 360 (-1043 - 196 x^2 + 30 x^4) \alpha^2 \mu^3 - 5045 040 (75 + 7 x^2) \alpha \mu^4 - \\
& 151 351 200 \mu^5) - 538 137 600 \alpha^2 \mu^2 (\alpha + 3 \mu)^3 (3 \mu + 2 \nu) + 40 040 T_f \alpha \mu \\
& (\alpha + 3 \mu)^2 (2520 \mu^2 (13 \mu + 7 \nu) - 56 \alpha \mu ((-225 + 8 x^2) \mu - 3 (105 + x^2) \nu) + \\
& \alpha^2 (-7 (306 - 256 x^2 + 21 x^4) \mu + 3 (1414 + 28 x^2 - 47 x^4) \nu))) + \\
& 120 120 \mu (\alpha - x^2 \alpha + 2 \mu) (\alpha + 3 \mu) (56 T_f x^4 \alpha^2 ((-5 + x^2) \alpha - 10 \mu) \mu (\alpha + 3 \mu) + \\
& 4 (\alpha + 3 \mu) (840 (-2 + x) \alpha ((-1 + x^2) \alpha - 2 \mu) \mu (\alpha + 3 \mu) + \\
& 7 T_f ((15 - 15 x^2 - 5 x^4 + x^6) \alpha^3) + 10 (-9 + 6 x^2 + x^4) \\
& \alpha^2 \mu + 60 (-3 + x^2) \alpha \mu^2 - 120 \mu^3)) (\mu + \nu) + \\
& x_1 (1680 T_f (-3 + x) \alpha \mu (\alpha + 2 \mu)^2 (\alpha + 3 \mu) + 3 T_f^2 (\alpha + 2 \mu) \\
& (117 \alpha^3 + 658 \alpha^2 \mu + 1260 \alpha \mu^2 + 840 \mu^3) - \\
& 1120 \alpha \mu (\alpha + 3 \mu)^2 ((3 + (-6 + x) x) \alpha \mu + 2 \alpha \nu + 2 \mu (5 \mu + 2 \nu))))), \\
& 1 / (1971 200 (\alpha + 3 \mu)^6) \times 27 T_f^4 \alpha^3 \mu (1320 T_f (2 - x) x^2 \alpha \mu (\alpha + 3 \mu) \\
& ((6 - 7 x^4 + x^6) \alpha - 14 (-1 + x^2) \mu) - \\
& 5280 T_f x_1 \alpha \mu (\alpha + 3 \mu) (x^7 \alpha - 7 x^5 (\alpha + 2 \mu) + 2 x^2 (3 \alpha + 7 \mu)) + \\
& x_2 (10 560 T_f (-1 + x^2) \alpha \mu (\alpha + 3 \mu) ((-6 - 6 x^2 + x^4) \alpha - 14 (1 + x^2) \mu) + \\
& T_f^2 ((-9530 + 10 197 x^4 - 165 x^6 - 550 x^8 + 48 x^{10}) \alpha^3 - \\
& 44 (1367 - 1407 x^4 + 15 x^6 + 25 x^8) \alpha^2 \mu - \\
& 660 (188 - 189 x^4 + x^6) \alpha \mu^2 + 83 160 (-1 + x^4) \mu^3) - \\
& 36 960 (-1 + x^4) \alpha \mu (\alpha + 3 \mu)^2 (3 \mu + 2 \nu))) \};
\end{aligned}$$

(*_ 8th order _*)

$$\begin{aligned}
\varphi_8 = & - (1 / (15 769 600 (\alpha + 3 \mu)^7)) 81 T_f^4 \alpha \mu \\
& (T_f^3 x_1 (76 975 \alpha^6 + 831 616 \alpha^5 \mu + 3 761 692 \alpha^4 \mu^2 + 9 192 480 \alpha^3 \mu^3 + 12 917 520 \alpha^2 \\
& \mu^4 + 9 979 200 \alpha \mu^5 + 3 326 400 \mu^6) + 44 T_f^2 (\alpha + 3 \mu) (-50 400 \mu^5 (\mu + \nu) + \\
& 25 200 \alpha \mu^4 ((-5 - 20 x + 9 x^2 + x^2) \mu + (-5 + x^2) \nu) + 1680 \alpha^2 \mu^3 \\
& (2 (-38 - 300 x + 135 x^2 + 15 x^4 - 5 x^6) \mu + (-76 + 30 x^2 + 5 x^4) \nu) + \\
& 40 \alpha^3 \mu^2 ((-1635 - 19 320 x + 8673 x^2 + 861 x^4 - 630 x^6 + 35 x^8) \mu + \\
& (-1635 + 966 x^2 + 315 x^4 - 7 x^6) \nu) + 10 \alpha^4 \mu \\
& ((-1689 - 26 760 x + 11 964 x^2 + 948 x^4 - 1148 x^6 + 140 x^8 - 15 x^{10}) \mu + \\
& (-1689 + 1338 x^2 + 623 x^4 - 28 x^6 - 15 x^8) \nu) + \\
& \alpha^5 ((-1755 - 35 100 x + 15 615 x^2 + 855 x^4 - 1620 x^6 + 350 x^8 - 75 x^{10} + 8 \\
& x^{10}) \mu + (-1755 + 1755 x^2 + 1020 x^4 - 70 x^6 - 75 x^8 + 8 x^{10}) \nu)) + \\
& 1760 T_f \alpha \mu (\alpha + 3 \mu)^2 (-840 \mu^3 (2 (-5 + 14 x) \mu + (-10 + 17 x) \nu) + \\
& 140 \alpha \mu^2 ((90 - 108 x - 120 x^2 + 23 x^3 - 30 x^5 + 24 x^7) \mu + \\
& 3 (30 - 51 x - 10 x^2 + 8 x^4) \nu) + 14 \alpha^2 \mu \\
& ((-1200 x^2 + 230 x^3 + 50 (-3 + x^2)^2 + x_1 (148 + 240 x^2 - 25 x^4)) \mu +
\end{aligned}$$

$$\begin{aligned}
& (-50 (-9 + 6 x^2 + x^4) + x^1 (-779 + 240 x^2 + 25 x^4)) \nu + \\
& \alpha^3 (35 (-120 x^1 + 23 x^3 - 2 (-15 + 15 x^2 - 5 x^4 + x^6) + x^1 (54 + \\
& \quad 24 x^2 - 5 x^4 + x^6)) \mu + (x^1 (-1881 + 840 x^2 + 175 x^4 - 35 x^6) + \\
& \quad 70 (15 - 15 x^2 - 5 x^4 + x^6)) \nu) + 123 200 \alpha \mu (\alpha + 3 \mu)^3 \\
& (8 \mu^2 (10 \mu^2 + 17 \mu \nu + 7 \nu^2) - 4 \alpha \mu ((28 - 160 x^1 + 45 x^1^2 + 15 x^2) \mu^2 + \\
& \quad (14 - 100 x^1 + 27 x^1^2 + 27 x^2) \mu \nu + 2 (-7 + 6 x^2) \nu^2) + \\
& \alpha^2 ((-76 - 40 x^1 + 5 x^1^2 + 66 x^2 + 19 x^4 + 6 x^1^2 (1 + 5 x^2) - \\
& \quad 8 x^1 (-32 + 15 x^2)) \mu^2 + (-62 + 42 x^2 + 23 x^4 - 40 x^1 (-5 + 3 x^2) + \\
& \quad 6 x^1^2 (-9 + 5 x^2)) \mu \nu + 2 (7 - 12 x^2 + 4 x^4) \nu^2));
\end{aligned}$$

$$\begin{aligned}
u8 = & \{1 / (5740134400 (\alpha + 3 \mu)^8) \times 81 \text{Tf}^5 \alpha \\
& \mu (-3003 \alpha (\text{Tf}^2 x^2 \alpha ((495 + 35 x^2 - 75 x^4 + 8 x^6) \alpha^3 + \\
& \quad 10 (308 + 14 x^2 - 15 x^4) \alpha^2 \mu + 140 (45 + x^2) \alpha \mu^2 + 4200 \mu^3) - \\
& \quad 40 \text{Tf} \mu (\alpha + 3 \mu) (3 (39 x^1 - 35 x^1 x^2 - 35 (-2 + x^1) x^4 + 7 (-2 + x^1) x^6) \alpha^3 + \\
& \quad 14 (30 x^4 + x^1 (47 - 15 x^2 (2 + x^2))) \alpha^2 \mu - \\
& \quad 420 x^1 (-3 + x^2) \alpha \mu^2 + 840 x^1 \mu^3) + \\
& \quad 2800 \mu (\alpha + 3 \mu)^2 (\mu ((-14 - 6 (-4 + x^1) x^1 + 6 (2 + (-4 + x^1) x^1) x^2 - x^4) \\
& \quad \alpha^2 - 4 (3 (-4 + x^1) x^1 + 2 (3 + x^2)) \alpha \mu + 8 \mu^2) + \\
& \quad 2 (\alpha + 2 \mu) (\alpha - 2 x^2 \alpha + 2 \mu) \nu) (3 \text{Tf} x^1 (\alpha + 2 \mu)^2 + \\
& \quad 4 (\alpha + 3 \mu) ((-1 + (-4 + x^1) x^1 + x^2) \alpha \mu + (-1 + x^2) \alpha \nu - 2 \mu (\mu + \nu))) + \\
& \quad 273 (\alpha - x^2 \alpha + 2 \mu) (\text{Tf}^3 x^1 (76975 \alpha^6 + 831616 \alpha^5 \mu + 3761692 \alpha^4 \mu^2 + \\
& \quad 9192480 \alpha^3 \mu^3 + 12917520 \alpha^2 \mu^4 + 9979200 \alpha \mu^5 + 3326400 \mu^6) + \\
& \quad 44 \text{Tf}^2 (\alpha + 3 \mu) (-50400 \mu^5 (\mu + \nu) + 25200 \alpha \mu^4 \\
& \quad ((-5 - 20 x^1 + 9 x^1^2 + x^2) \mu + (-5 + x^2) \nu) + 1680 \alpha^2 \mu^3 \\
& \quad (2 (-38 - 300 x^1 + 135 x^1^2 + 15 x^2 - 5 x^4) \mu + (-76 + 30 x^2 + 5 x^4) \nu) + \\
& \quad 40 \alpha^3 \mu^2 ((-1635 - 19320 x^1 + 8673 x^1^2 + 861 x^2 - 630 x^4 + 35 x^6) \\
& \quad \mu + (-1635 + 966 x^2 + 315 x^4 - 7 x^6) \nu) + 10 \alpha^4 \mu \\
& \quad ((-1689 - 26760 x^1 + 11964 x^1^2 + 948 x^2 - 1148 x^4 + 140 x^6 - 15 x^8) \mu + \\
& \quad (-1689 + 1338 x^2 + 623 x^4 - 28 x^6 - 15 x^8) \nu) + \alpha^5 ((-1755 - 35100 \\
& \quad x^1 + 15615 x^1^2 + 855 x^2 - 1620 x^4 + 350 x^6 - 75 x^8 + 6 x^{10}) \mu + \\
& \quad (-1755 + 1755 x^2 + 1020 x^4 - 70 x^6 - 75 x^8 + 8 x^{10}) \nu) + \\
& \quad 1760 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 (-840 \mu^3 (2 (-5 + 14 x^1) \mu + (-10 + 17 x^1) \nu) + \\
& \quad 140 \alpha \mu^2 ((90 - 108 x^1 - 120 x^1^2 + 23 x^1^3 - 30 x^2 + 24 x^1 x^2) \\
& \quad \mu + 3 (30 - 51 x^1 - 10 x^2 + 8 x^1 x^2) \nu) + 14 \alpha^2 \mu \\
& \quad ((-1200 x^1^2 + 230 x^1^3 + 50 (-3 + x^2)^2 + x^1 (148 + 240 x^2 - 25 x^4)) \\
& \quad \mu + (-50 (-9 + 6 x^2 + x^4) + x^1 (-779 + 240 x^2 + 25 x^4)) \nu) + \\
& \quad \alpha^3 (35 (-120 x^1 + 23 x^3 - 2 (-15 + 15 x^2 - 5 x^4 + x^6) + x^1 (54 + 24 x^2 \\
& \quad - 5 x^4 + x^6)) \mu + (x^1 (-1881 + 840 x^2 + 175 x^4 - 35 x^6) + \\
& \quad 70 (15 - 15 x^2 - 5 x^4 + x^6)) \nu) + 123 200 \alpha \mu (\alpha + 3 \mu)^3 \\
& (8 \mu^2 (10 \mu^2 + 17 \mu \nu + 7 \nu^2) - 4 \alpha \mu ((28 - 160 x^1 + 45 x^1^2 + 15 x^2) \mu^2 + \\
& \quad \mu^2 + (14 - 100 x^1 + 27 x^1^2 + 27 x^2) \mu \nu + 2 (-7 + 6 x^2) \nu^2) + \\
& \alpha^2 ((-76 - 40 x^1 + 5 x^1^2 + 66 x^2 + 19 x^4 + 6 x^1^2 (1 + 5 x^2) - 8 \\
& \quad x^1 (-32 + 15 x^2)) \mu^2 + (-62 + 42 x^2 + 23 x^4 - 40 x^1 (-5 + 3 x^2) + \\
& \quad 6 x^1^2 (-9 + 5 x^2)) \mu \nu + 2 (7 - 12 x^2 + 4 x^4) \nu^2)) - \\
& \quad 3003 \alpha (\text{Tf} x^2 \alpha ((-5 + x^2) \alpha - 10 \mu) + 40 (-2 + x^1) \mu (\alpha - x^2 \alpha + 2 \mu) (\alpha + 3 \mu)) \\
& (-56 \text{Tf} x^2 \alpha^2 ((-5 + x^2) \alpha - 10 \mu) \mu (\alpha + 3 \mu) - \\
& \quad 4 (\alpha + 3 \mu) (840 (-2 + x^1) \alpha ((-1 + x^2) \alpha - 2 \mu) \mu (\alpha + 3 \mu) + \\
& \quad 7 \text{Tf} (- (15 - 15 x^2 - 5 x^4 + x^6) \alpha^3) + 10 (-9 + 6 x^2 + x^4) \\
& \quad \alpha^2 \mu + 60 (-3 + x^2) \alpha \mu^2 - 120 \mu^3)) (\mu + \nu) + \\
& \quad x^1 (-1680 \text{Tf} (-3 + x^1) \alpha \mu (\alpha + 2 \mu)^2 (\alpha + 3 \mu) - 3 \text{Tf}^2 (\alpha + 2 \mu)
\end{aligned}$$

$$\begin{aligned}
& (117 \alpha^3 + 658 \alpha^2 \mu + 1260 \alpha \mu^2 + 840 \mu^3) + \\
& 1120 \alpha \mu (\alpha + 3 \mu)^2 (-((3 + (-6 + x1) x1) \alpha \mu) + 2 \alpha \nu + 2 \mu (5 \mu + 2 \nu))) - \\
& (2 - x1) \alpha (3843840 \text{Tf} x1^2 \alpha^3 \mu^2 (\alpha + 3 \mu)^2 ((6 - 35 x2^4 + 7 x2^6) \alpha + \\
& 14 (1 - 5 x2^4) \mu) - 1456 x1 \alpha^2 \mu (\alpha + 3 \mu) \\
& (10560 \text{Tf} \alpha \mu (\alpha + 3 \mu) ((6 - 35 x2^4 + 7 x2^6) \alpha + 14 (1 - 5 x2^4) \mu) + \\
& \text{Tf}^2 ((-9530 + 50985 x2^4 - 1155 x2^6 - 4950 x2^8 + 528 x2^{10}) \\
& \alpha^3 - 44 (1367 - 7035 x2^4 + 105 x2^6 + 225 x2^8) \alpha^2 \mu - 660 \\
& (188 - 945 x2^4 + 7 x2^6) \alpha \mu^2 + 83160 (-1 + 5 x2^4) \mu^3) - \\
& 36960 (-1 + 5 x2^4) \alpha \mu (\alpha + 3 \mu)^2 (3 \mu + 2 \nu)) + 24024 (2 - x1) \alpha \\
& \mu (\alpha + 3 \mu) (\text{Tf}^2 x2^4 \alpha ((495 + 35 x2^2 - 75 x2^4 + 8 x2^6) \alpha^3 + 10 \\
& (308 + 14 x2^2 - 15 x2^4) \alpha^2 \mu + 140 (45 + x2^2) \alpha \mu^2 + 4200 \mu^3) - \\
& 40 \text{Tf} \mu (\alpha + 3 \mu) (3 (39 x1 - 35 x1 x2^2 - 35 (-2 + x1) x2^4 + 7 (-2 + x1) x2^6) \\
& \alpha^3 + 14 (30 x2^4 + x1 (47 - 15 x2^2 (2 + x2^2))) \alpha^2 \mu - 420 \\
& x1 (-3 + x2^2) \alpha \mu^2 + 840 x1 \mu^3) + 2800 \mu (\alpha + 3 \mu)^2 \\
& (\mu ((-14 - 6 (-4 + x1) x1 + 6 (2 + (-4 + x1) x1) x2^2 - x2^4) \alpha^2 - 4 (3 (-4 + x1) \\
& x1 + 2 (3 + x2^2)) \alpha \mu + 8 \mu^2) + 2 (\alpha + 2 \mu) (\alpha - 2 x2^2 \alpha + 2 \mu) \nu)) + \\
& 840840 \alpha \mu (\alpha + 3 \mu) (\text{Tf} x2^4 \alpha ((-5 + x2^2) \alpha - 10 \mu) + 40 (-2 + x1) \mu \\
& (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu)) (3 \text{Tf} x1 (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) ((-1 + (-4 + x1) x1 + x2^2) \alpha \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu))) + \\
& x2^4 \alpha (192192 \text{Tf}^2 \alpha \mu (\alpha + 3 \mu) ((495 + 35 x2^2 - 75 x2^4 + 8 x2^6) \alpha^3 + \\
& 10 (308 + 14 x2^2 - 15 x2^4) \alpha^2 \mu + 140 (45 + x2^2) \alpha \mu^2 + 4200 \mu^3) + \\
& \text{Tf}^3 ((-4403035 - 1918917 x2^2 + 1301157 x2^4 + 101101 x2^6 - 70616 x2^8 + \\
& 5067 x2^{10}) \alpha^5 - 91 (497222 + 180312 x2^2 - 87813 x2^4 - 4444 x2^6 + \\
& 1552 x2^8) \alpha^4 \mu + 4004 (-46230 - 12894 x2^2 + 4050 x2^4 + 101 x2^6) \\
& \alpha^3 \mu^2 + 360360 (-1043 - 196 x2^2 + 30 x2^4) \alpha^2 \\
& \mu^3 - 5045040 (75 + 7 x2^2) \alpha \mu^4 - 151351200 \mu^5) - \\
& 538137600 \alpha^2 \mu^2 (\alpha + 3 \mu)^3 (3 \mu + 2 \nu) + 40040 \text{Tf} \alpha \mu (\alpha + 3 \mu)^2 \\
& (2520 \mu^2 (13 \mu + 7 \nu) - 56 \alpha \mu ((-225 + 8 x2^2) \mu - 3 (105 + x2^2) \nu) + \alpha^2 \\
& (-7 (306 - 256 x2^2 + 21 x2^4) \mu + 3 (1414 + 28 x2^2 - 47 x2^4) \nu))) + \\
& 120120 \mu (\alpha - x2^2 \alpha + 2 \mu) (\alpha + 3 \mu) (56 \text{Tf} x2^4 \alpha^2 ((-5 + x2^2) \alpha - 10 \mu) \\
& \mu (\alpha + 3 \mu) + 4 (\alpha + 3 \mu) (840 (-2 + x1) \alpha ((-1 + x2^2) \alpha - 2 \mu) \\
& \mu (\alpha + 3 \mu) + 7 \text{Tf} (-((15 - 15 x2^2 - 5 x2^4 + x2^6) \alpha^3) + \\
& 10 (-9 + 6 x2^2 + x2^4) \alpha^2 \mu + 60 (-3 + x2^2) \alpha \mu^2 - 120 \mu^3)) (\mu + \nu) + \\
& x1 (1680 \text{Tf} (-3 + x1) \alpha \mu (\alpha + 2 \mu)^2 (\alpha + 3 \mu) + 3 \text{Tf}^2 (\alpha + 2 \mu) \\
& (117 \alpha^3 + 658 \alpha^2 \mu + 1260 \alpha \mu^2 + 840 \mu^3) - 1120 \alpha \mu (\alpha + 3 \mu)^2 \\
& (-((3 + (-6 + x1) x1) \alpha \mu) + 2 \alpha \nu + 2 \mu (5 \mu + 2 \nu))))), \\
& 1 / (3942400 (\alpha + 3 \mu)^7) \times 81 \text{Tf}^5 \alpha^4 \mu^2 (-165 \text{Tf} x2 \\
& ((6 - 7 x2^4 + x2^6) \alpha - 14 (-1 + x2^4) \mu) \\
& (3 \text{Tf} x1 (\alpha + 2 \mu)^2 + \\
& 4 (\alpha + 3 \mu) ((-1 + (-4 + x1) x1 + x2^2) \alpha \mu + (-1 + x2^2) \alpha \nu - 2 \mu (\mu + \nu))) - \\
& (2 - x1) (1320 \text{Tf} (2 - x1) x2 \alpha \mu (\alpha + 3 \mu) ((6 - 7 x2^4 + x2^6) \alpha - 14 (-1 + x2^4) \mu) - \\
& 5280 \text{Tf} x1 \alpha \mu (\alpha + 3 \mu) (x2^7 \alpha - 7 x2^5 (\alpha + 2 \mu) + 2 x2 (3 \alpha + 7 \mu)) + \\
& x2 (10560 \text{Tf} (-1 + x2^2) \alpha \mu (\alpha + 3 \mu) ((-6 - 6 x2^2 + x2^4) \alpha - 14 (1 + x2^2) \mu) + \\
& \text{Tf}^2 ((-9530 + 10197 x2^4 - 165 x2^6 - 550 x2^8 + 48 x2^{10}) \alpha^3 - 44 (1367 - \\
& 1407 x2^4 + 15 x2^6 + 25 x2^8) \alpha^2 \mu - 660 (188 - 189 x2^4 + x2^6) \alpha \mu^2 + \\
& 83160 (-1 + x2^4) \mu^3) - 36960 (-1 + x2^4) \alpha \mu (\alpha + 3 \mu)^2 (3 \mu + 2 \nu)))));
\end{aligned}$$

(=====*)

(*Verification of equations 1-8*)

(=====*)

```

(*Equation 1*)
eq1 = Grad[φ2, {x1, x2}] - μ Laplacian[u0, {x1, x2}] // FullSimplify;
test1 = (eq1 == {0, 0});

(*Equation 2*)
eq2 = Div[u2, {x1, x2}] + Grad[φ2, {x1, x2}].u0 // FullSimplify;
test2 = (eq2 == 0);

(*Equation 3*)
eq3 = Grad[φ4, {x1, x2}] -
      μ Laplacian[u2, {x1, x2}] - (μ + ν) Grad[Div[u2, {x1, x2}], {x1, x2}] +
      Grad[u0, {x1, x2}].u2 + Grad[u2, {x1, x2}].u0 // FullSimplify;
test3 = (eq3 == {0, 0});

(*Equation 4*)
eq4 = Div[u4, {x1, x2}] + Grad[φ2, {x1, x2}].u2 +
      Grad[φ4, {x1, x2}].u0 + φ2 Div[u2, {x1, x2}] // FullSimplify;
test4 = (eq4 == 0);

(*Equation 5*)
eq5 = Grad[φ6, {x1, x2}] -
      μ Laplacian[u4, {x1, x2}] - (μ + ν) Grad[Div[u4, {x1, x2}], {x1, x2}] +
      Grad[u0, {x1, x2}].u4 + Grad[u2, {x1, x2}].u2 + Grad[u4, {x1, x2}].u0 +
      φ2 Grad[u0, {x1, x2}].u2 + φ2 Grad[u2, {x1, x2}].u0 // FullSimplify;
test5 = (eq5 == {0, 0});

(*Equation 6*)
eq6 = Div[u6, {x1, x2}] + Grad[φ2, {x1, x2}].u4 + Grad[φ4, {x1, x2}].u2 +
      Grad[φ6, {x1, x2}].u0 + φ4 Div[u2, {x1, x2}] + φ2 Div[u4, {x1, x2}] // FullSimplify;
test6 = (eq6 == 0);

(*Equation 7-momentum for φ8*)
eq7 = Grad[φ8, {x1, x2}] - μ Laplacian[u6, {x1, x2}] - (μ + ν)
      Grad[Div[u6, {x1, x2}], {x1, x2}] + (Grad[u6, {x1, x2}].u0 + Grad[u0, {x1, x2}].u6) +
      (Grad[u4, {x1, x2}].u2 + Grad[u2, {x1, x2}].u4) +
      φ2 (Grad[u4, {x1, x2}].u0 + Grad[u2, {x1, x2}].u2 + Grad[u0, {x1, x2}].u4) +
      φ4 (Grad[u2, {x1, x2}].u0 + Grad[u0, {x1, x2}].u2) // FullSimplify;
test7 = (eq7 == {0, 0});

(*Equation 8-continuity for u8*)
eq8 = Div[u8, {x1, x2}] + Grad[φ2, {x1, x2}].u6 + Grad[φ4, {x1, x2}].u4 +
      Grad[φ6, {x1, x2}].u2 + Grad[φ8, {x1, x2}].u0 + φ6 Div[u2, {x1, x2}] +
      φ4 Div[u4, {x1, x2}] + φ2 Div[u6, {x1, x2}] // FullSimplify;
test8 = (eq8 == 0);

(*_ Boundary checks: v=0 at x2=±1 _*)
b2 = {FullSimplify[u2[[2]] /. x2 → 1] == 0, FullSimplify[u2[[2]] /. x2 → -1] == 0};
b4 = {FullSimplify[u4[[2]] /. x2 → 1] == 0, FullSimplify[u4[[2]] /. x2 → -1] == 0};
b6 = {FullSimplify[u6[[2]] /. x2 → 1] == 0, FullSimplify[u6[[2]] /. x2 → -1] == 0};

```

```
b8 = {FullSimplify[u8[[2]] /. x2 → 1] == 0, FullSimplify[u8[[2]] /. x2 → -1] == 0};
allBdry = And @@ Flatten[{b2, b4, b6, b8}];
```

```
(*=====*)
(*Output section*)
(*=====*)
Print["====="];
Print["Verification up to 8th order (x1,x2)"];
Print["====="];
Print["eq1: ", test1];
Print["eq2: ", test2];
Print["eq3: ", test3];
Print["eq4: ", test4];
Print["eq5: ", test5];
Print["eq6: ", test6];
Print["eq7: ", test7];
Print["eq8: ", test8];
Print["All main equations passed: ",
      test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8];

Print["-----"];
Print["Boundary conditions (v = 0 at x2 = ±1):"];
Print[" u2: ", b2];
Print[" u4: ", b4];
Print[" u6: ", b6];
Print[" u8: ", b8];
Print["All boundary tests passed: ", allBdry];

Print["-----"];
Print["Overall status: ",
      test1 && test2 && test3 && test4 && test5 && test6 && test7 && test8 && allBdry];
Print["====="];
```

```
Total time: 0.261031 s
```

```
=====
```

```
Verification up to 8th order (x1,x2)
```

```
=====
```

```
eq1: True
```

```
eq2: True
```

```
eq3: True
```

```
eq4: True
```

```
eq5: True
```

```
eq6: True
```

```
eq7: True
```

```
eq8: True
```

```
All main equations passed: True
```

Boundary conditions ($v = 0$ at $x_2 = \pm 1$):

u2: {True, True}

u4: {True, True}

u6: {True, True}

u8: {True, True}

All boundary tests passed: True

Overall status: True
=====